Newton – Euler Equation

Euler 1st law (translation) $\frac{d}{dt}{}^I\vec{P}_{G/O} = m_G\,{}^I\vec{a}_{G/O} = \vec{F}_{ext}$

$${}^I\vec{P}_{G/O} = m_G\,{}^I\vec{v}_{G/O} = m_G \frac{1}{m_G}\sum_i m_i\,{}^I\vec{v}_{i/O}$$

$$\frac{d}{dt}{}^I\vec{P}_{G/O} = m_G \frac{1}{m_G}\sum_i m_i\,{}^I\vec{a}_{i/O} = \sum_i {}^I\vec{F}_i = \sum_i \left[{}^I\vec{F}_{i,ext} + \sum_j {}^I\vec{F}_{i,j}\right] = F_{ext}$$

Euler 2nd law (rotation) $\frac{d}{dt}{}^I\vec{h}_{S/G} = \sum_i {}^I\vec{r}_{i/G} \times {}^I\vec{F}_{i,ext} = {}^I\vec{M}_{G,ext}$

$$\begin{aligned}
\frac{d}{dt}{}^I\vec{h}_{S/G} &= \frac{d}{dt}\left[\sum_i m_i\,{}^I\vec{r}_{i/G} \times {}^I\vec{v}_{i/G}\right] \\
&= \sum_i m_i\,{}^I\vec{r}_{i/G} \times {}^I\vec{a}_{i/G} \\
&= \sum_i m_i\,{}^I\vec{r}_{i/G} \times \left[{}^I\vec{a}_{i/O} - {}^I\vec{a}_{G/O}\right] \\
&= \sum_i {}^I\vec{r}_{i/G} \times {}^I\vec{F}_i \\
&= \sum_i {}^I\vec{r}_{i/G} \times \left[{}^I\vec{F}_{i,ext} + \sum_j {}^I\vec{F}_{i,j}\right] \\
&= \sum_i {}^I\vec{r}_{i/G} \times {}^I\vec{F}_{i,ext} + \underbrace{\sum_i {}^I\vec{r}_{i/G} \times \sum_j {}^I\vec{F}_{i,j}}_{0 \text{ under internal moment assumption}} \\
&= \sum_i {}^I\vec{r}_{i/G} \times {}^I\vec{F}_{i,ext}
\end{aligned}$$

Newton Euler Equation: $({}^I\vec{M}_{G,ext})_{/B} = {}^BI_G \frac{d}{dt}\omega_{B/B} + \omega_{B/B} \times {}^BI_G\,\omega_{B/B}$

$$\begin{aligned}
{}^I\vec{M}_{G,ext} &= \frac{d}{dt}{}^I\vec{h}_{S/G} \\
&= \left[\frac{d}{dt}({}^I\vec{h}_{S/G})_{/B}\right]_{/I} + \omega_{B/I} \times {}^I\vec{h}_{S/G} \\
&= \left[\frac{d}{dt}\left({}^BI_G\,\omega_{B/B}\right)\right]_{/I} + \omega_{B/I} \times \left({}^BI_G\,\omega_{B/B}\right)_{/I} \\
&= \left[{}^BI_G \cdot \frac{d}{dt}\omega_{B/B}\right]_{/I} + \left[\omega_{B/B} \times {}^BI_G\,\omega_{B/B}\right]_{/I} \\
&= \left[{}^BI_G \frac{d}{dt}\omega_{B/B} + \omega_{B/B} \times {}^BI_G\,\omega_{B/B}\right]_{/I}
\end{aligned}$$

Newton – Euler Dynamics in SE(3)

translation in inertial frame: $m_G\,{}^I\vec{a}_{G/O} = F_{ext}$

rotation in body frame: ${}^BI_G \frac{d}{dt}\omega_{B/B} + \omega_{B/B} \times {}^BI_G\,\omega_{B/B} = ({}^I\vec{M}_{G,ext})_{/B}$

$$X = \begin{bmatrix} {}^I r_{G/O} \\ q_{B/I} \\ ({}^I v_{G/O})_{/B} \\ \omega_{B/B} \end{bmatrix}$$

- position, described in I frame
- orientation described in I frame
- inertial velocity, described in B frame
- angular velocity, described in B frame

1. $\frac{d}{dt}{}^{I}\vec{r}_{G/O} = {}^{I}\vec{V}_{G/O} = q_{B/I}\left[{}^{I}\vec{V}_{G/O}\right]_{B} = q_{B/I} * \begin{bmatrix} 0 \\ [{}^{I}\vec{V}_{G/O}]_{B} \end{bmatrix} * q_{B/I}^{-1}$

2. $\frac{d}{dt} q_{B/I} = q_{B/I} * \begin{bmatrix} 0 \\ \frac{1}{2}\omega_{B/B} \end{bmatrix}$

3. $\frac{d}{dt}\left[{}^{I}\vec{V}_{G/O}\right]_{I} = \left[\frac{d}{dt}\left({}^{I}\vec{V}_{G/O}\right)_{B}\right]_{I} + \omega_{B/I} \times {}^{I}\vec{V}_{G/O}$

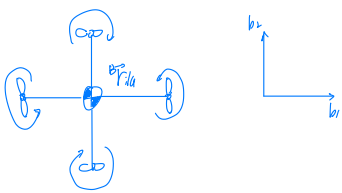
$\frac{d}{dt}\left[{}^{I}\vec{V}_{G/O}\right]_{B} = \left[{}^{I}\vec{a}_{G/O} - \omega_{B/I} \times {}^{I}\vec{V}_{G/O}\right]_{B}$

$= \frac{1}{m_G}{}^{B}\vec{F}_{ext} - \omega_{B/B} \times \left({}^{I}\vec{V}_{G/O}\right)_{B}$

4. $\frac{d}{dt}\left(\omega_{B/B}\right) = {}^{B}I_{G}^{-1}\left[\left({}^{I}M_{G,ext}\right)_{B} - \omega_{B/B} \times {}^{B}I_{G}\,\omega_{B/B}\right]$

$$X = \begin{bmatrix} {}^{I}r_{G/O} \\ q_{B/I} \\ \left({}^{I}V_{G/O}\right)_{B} \\ \omega_{B/B} \end{bmatrix} \qquad \dot{X} = \begin{bmatrix} q_{B/I} * \begin{bmatrix} 0 \\ [{}^{I}\vec{V}_{G/O}]_{B} \end{bmatrix} * q_{B/I}^{-1} \\ q_{B/I} * \begin{bmatrix} 0 \\ \frac{1}{2}\omega_{B/B} \end{bmatrix} \\ \frac{1}{m_G}{}^{B}\vec{F}_{ext} - \omega_{B/B} \times \left({}^{I}\vec{V}_{G/O}\right)_{B} \\ {}^{B}I_{G}^{-1}\left[\left({}^{I}M_{G,ext}\right)_{B} - \omega_{B/B} \times {}^{B}I_{G}\,\omega_{B/B}\right] \end{bmatrix}$$

Quadrotor



each propeller exerts a force (thrust) $F_i = k_T \omega_i^2$ in its direction

a torque $T_i = k_M \omega_i^2$ in this direction

ω_i is the angular velocity of each propeller

$${}^{B}\vec{F}_{ext} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ k_T & k_T & k_T & k_T \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} + Q^{T} \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix}$$

$$\left({}^{I}M_{G,ext}\right)_{B} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ k_M & -k_M & k_M & -k_M \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} + \sum_{i} {}^{B}\vec{r}_{i/G} \times \begin{bmatrix} 0 \\ 0 \\ k_T \end{bmatrix} \omega_i^2$$